

Homework 02 - Operators And Decision Making

Topics Covered

- Variables
- Operators
- `if`
- `if-else`
- `else-if`
- nested `if`
- `switch`

Instructions For Students

1. Write answers in your notebook first.
2. Then write Java programs in your IDE.
3. Use meaningful variable names.
4. Mention the output for every output question.
5. Submit `.java` files for coding questions.

Part A: Short Answer Questions

Answer in one or two lines.

1. What is an operator?
2. What is an operand?
3. What is the difference between `=` and `==` ?
4. What does `%` operator return?
5. What is the difference between `&&` and `||` ?
6. What does `!` operator do?
7. What is the difference between `a++` and `++a` ?
8. What is a control statement?
9. What is a condition?
10. When do we use `if-else` ?
11. When do we use `else-if` ?
12. When do we use `switch` ?
13. Why do we use `break` inside `switch` ?
14. What is the use of `default` in `switch` ?
15. What is nested `if` ?

Part B: Find The Output

Question 1

```
public class OutputOne {  
    public static void main(String[] args) {  
        int a = 10;  
        int b = 3;  
  
        System.out.println(a / b);  
        System.out.println(a % b);  
    }  
}
```

Question 2

```
public class OutputTwo {  
    public static void main(String[] args) {  
        System.out.println(10 + 20);  
        System.out.println("10" + 20);  
        System.out.println("Sum: " + 10 + 20);  
        System.out.println("Sum: " + (10 + 20));  
    }  
}
```

Question 3

```
public class OutputThree {  
    public static void main(String[] args) {  
        int a = 5;  
        int b = a++;  
  
        System.out.println(a);  
        System.out.println(b);  
    }  
}
```

Question 4

```
public class OutputFour {  
    public static void main(String[] args) {  
        int a = 5;  
        int b = ++a;  
  
        System.out.println(a);  
        System.out.println(b);  
    }  
}
```

Question 5

```
public class OutputFive {
    public static void main(String[] args) {
        int a = 10;

        a += 5;
        a *= 2;
        a -= 10;

        System.out.println(a);
    }
}
```

Question 6

```
public class OutputSix {
    public static void main(String[] args) {
        int marks = 35;

        if (marks >= 40) {
            System.out.println("Pass");
        } else {
            System.out.println("Fail");
        }
    }
}
```

Question 7

```
public class OutputSeven {
    public static void main(String[] args) {
        int number = 11;

        if (number % 2 == 0) {
            System.out.println("Even");
        } else {
            System.out.println("Odd");
        }
    }
}
```

Question 8

```
public class OutputEight {
    public static void main(String[] args) {
        int marks = 82;

        if (marks >= 90) {
            System.out.println("A+");
        } else if (marks >= 75) {
            System.out.println("A");
        } else if (marks >= 60) {
            System.out.println("B");
        } else if (marks >= 40) {
            System.out.println("C");
        } else {
            System.out.println("Fail");
        }
    }
}
```

Question 9

```
public class OutputNine {
    public static void main(String[] args) {
        int day = 3;

        switch (day) {
            case 1:
                System.out.println("Monday");
                break;
            case 2:
                System.out.println("Tuesday");
                break;
            case 3:
                System.out.println("Wednesday");
                break;
            default:
                System.out.println("Invalid day");
        }
    }
}
```

Question 10

```
public class OutputTen {
    public static void main(String[] args) {
        int day = 2;

        switch (day) {
            case 1:
                System.out.println("Monday");
            case 2:
                System.out.println("Tuesday");
            case 3:
                System.out.println("Wednesday");
            default:
                System.out.println("Invalid day");
        }
    }
}
```

Part C: Find And Correct The Error

Question 1

```
int age == 20;
```

Question 2

```
if (age >= 18);
{
    System.out.println("Eligible");
}
```

Question 3

```
int marks = 50;

if marks >= 40 {
    System.out.println("Pass");
}
```

Question 4

```
int day = 1;

switch day {
    case 1:
        System.out.println("Monday");
        break;
}
```

Question 5

```
int number = 10;

if (number % 2 == 0) {
    System.out.println("Even");
}
```

Part D: Coding Practice

Write Java programs for the following.

Program 1: Simple Calculator

Create two integer variables and print:

- addition
- subtraction
- multiplication
- division
- remainder

Program 2: Even Or Odd

Create one integer variable and check whether it is even or odd.

Program 3: Voting Eligibility

Create an `age` variable.

If age is greater than or equal to 18, print:

```
Eligible to vote
```

Otherwise print:

```
Not eligible to vote
```

Program 4: Pass Or Fail

Create a `marks` variable.

If marks are greater than or equal to 40, print `Pass`, otherwise print `Fail`.

Program 5: Grade Calculator

Create a `marks` variable and print grade:

Marks	Grade
<code>>= 90</code>	A+
<code>>= 75</code>	A
<code>>= 60</code>	B
<code>>= 40</code>	C
<code>< 40</code>	Fail

Program 6: Positive, Negative, Or Zero

Create one integer variable.

Print:

- `Positive number`
- `Negative number`
- `Zero`

Program 7: Largest Of Two Numbers

Create two integer variables and print which number is greater.

If both numbers are equal, print:

```
Both numbers are equal
```

Program 8: Largest Of Three Numbers

Create three integer variables and find the largest number.

Program 9: Login Check

Create:

```
String username = "admin";
String password = "1234";
```

If both are correct, print:

```
Login successful
```

Otherwise print:

```
Invalid login
```

Note: For now, you may use `==`. Later we will learn `.equals()` for String comparison.

Program 10: Discount Eligibility

Create:

```
boolean isStudent = true;
boolean isSeniorCitizen = false;
```

If student or senior citizen, print:

```
Discount available
```

Otherwise print:

```
No discount
```

Program 11: Day Name Using Switch

Create an integer variable `day` .

Print day name:

Value	Day
1	Monday
2	Tuesday
3	Wednesday
4	Thursday
5	Friday
6	Saturday
7	Sunday

For any other value, print:

```
Invalid day
```

Program 12: Simple Menu Using Switch

Create an integer variable `choice` .

Print:

Choice	Message
1	Add money
2	Withdraw money
3	Check balance
4	Exit

For any other choice, print:

Invalid choice

Part E: Slightly Challenging Questions

Challenge 1: Electricity Bill

Create variable:

```
int units = 180;
```

Calculate bill:

- First 100 units: 5 rupees per unit
- Next units: 8 rupees per unit

Example:

For 180 units:

```
100 * 5 + 80 * 8
```

Challenge 2: Simple ATM Withdrawal

Create:

```
int balance = 5000;  
int withdrawAmount = 2000;
```

If withdraw amount is less than or equal to balance, print:

```
Withdrawal successful  
Remaining balance: 3000
```

Otherwise print:

```
Insufficient balance
```

Challenge 3: Number Divisible By 3 And 5

Create one integer variable.

Check:

- divisible by both 3 and 5
- divisible only by 3
- divisible only by 5
- not divisible by 3 or 5

Challenge 4: Simple Calculator Using Switch

Create:

```
int a = 20;
int b = 5;
char operator = '+';
```

Use `switch` on `operator` .

Support:

- `+`
- `-`
- `*`
- `/`
- `%`

Part F: Oral Revision Questions

Teacher can ask these in next class:

1. What does `%` return?
2. What is output of `10 / 3` ?
3. What is difference between `=` and `==` ?
4. What does `&&` mean?
5. What does `||` mean?
6. What happens if `if` condition is false?
7. Why do we use `else-if` ?
8. Why do we use `break` in `switch` ?
9. What happens if `break` is missing in switch?
10. When should we use `switch` instead of `if-else` ?